INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April – May, 2025

Degree: **B.Tech**, Semester: **Fourth**

Stream: **CSE(AI & ML)/ CSE(IOT, CS, BT)/ CSE/CSIT/CST/CSE(AI)/CSE(IoT)/ CSBS/IT**

Paper Code: **PCCCS405**

Paper Name: **Advanced Programming (OOP)**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1.	Why Java is called Architectural?	2
2.	Explain each term of the following: public static void main (String [] args).	2
3.	How does JavaScript differ from Java, and what is their relationship in web development?	2
4.	Write a program to print a string in reverse order.	2
5.	Is it possible to call one constructor from another in a class with multiple constructors? If so, how is it achieved in Java?	2
6.	Compare and contrast overloading and overriding methods.	2
7.	Explain the syntax of BufferedReader object creation.	2
8.	How does thread scheduling work in Java, and what factors influence the execution order of threads?	2
9.	In multi-threading how can we ensure that a resource isn't used by multiple threads simultaneously?	2
10.	What are three concrete implementations of the List interface in Java, and how do they differ in functionality?	2
11.	Define the Hierarchy of Collection Framework in Java.	2

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12. What is a Model? 2

13. What are the primary goals of UML (Unified Modeling Language), and how do they contribute to software development? 2

14. Explain the purpose and application of the AWT (Abstract Window Toolkit) package in Java, highlighting its usage and relevance within the Java programming environment. 2

15. Explain the different lifecycle methods of an Android Activity and their role in handling state changes. 2

Part - B

Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Develop a diagram illustrating the process of compiling and interpreting Java code, highlighting the role of the Just-In-Time Compiler (JIT) and its significance in this process. 5

17. Develop a Java application to generate Electricity bills. Create a class with the following members: Consumer no., consumer name, previous month reading, current month reading, and type of EB connection (i.e., domestic or commercial). Compute the bill amount using the following tariff. If the type of the EB connection is domestic, calculate the amount to be paid as follows: 5
First 100 units - Rs. 1 per unit

101-200 units - Rs. 2.50 per unit

201 -500 units - Rs. 4 per unit

> 501 units - Rs. 6 per unit

18. Explain the different types of operators in Java with suitable examples for each. 5

19. Evaluate the role of abstract classes in Java inheritance. Analyse the advantages and limitations of using abstract classes compared to interfaces in object-oriented design. 5

20. Explain yield() and join() with proper code snippet. 5

21. Implement a java program to remove all the repeated element from the ArrayList. 5

22. Demonstrate the use of annotations in Java by writing a program and explaining its functionality with examples. 5

23. Explain internal structure diagram with a suitable example. 5

24. Difference between Swing and AWT. Write a Swing Program to use TextField, Button. 5

Part - C

Attempt any 5 questions out of 8 questions

Each question carries 10 Marks (10 × 5)

25. Describe the architecture of the Java Virtual Machine (JVM) with a detailed diagram, explaining the functionality of each component. **10**

26. a) Describe broadcast receivers. How is it implemented? **10**
b) Explain each token in the statement: System.out.println("Hello Java");

27. Design an abstract class shape having two methods calculate_area() and display(). Create rectangle, triangle, and ellipse classes by inheriting the shape class and overriding the above methods to suitably implement rectangle, triangle, and ellipse classes. **10**

28. What is array of objects? Describe the use of Access Modifiers in Java? Draw a table showing the scope of accessibility? Justify: **10**
  a) Why we declare main () method as public and static member?
  b) Why we generally declare constructor as public member?
  c) Why there is no destructor in java?

29. Explain the concept of multithreading in Java, highlighting its key features and importance. Write a Java program to demonstrate the creation and execution of multiple threads using both the Thread class and the Runnable interface. Additionally, discuss the advantages of multithreading in Java applications and how it improves performance and resource utilization. **10**

30. Write a Java program that implements a method to take a HashSet as input and remove all duplicate elements. Explain the working of HashSet in preventing duplicates and discuss its advantages over other collection types. Additionally, demonstrate the program with sample input and output. **10**

31. The Library Loan System manages the loans and circulation of library resource material such as books, journals and audiovisuals. Students and faculty enjoy membership at the library. During checkout, the LLS obtains the student or faculty ID. Resource materials can be checked out (i.e. borrowed) for a duration of 2 weeks. A fine of 1 rupee a day is imposed on overdue materials. Draw a detailed Class diagram. **10**

32. Develop a Java GUI application using the Swing package that includes a Text Field, Button, and Label. Implement an event-driven approach where clicking the button retrieves the text entered in the text field and displays it in the label. Ensure proper layout management and demonstrate the functionality with a complete Java program, including sample input and output **10**

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